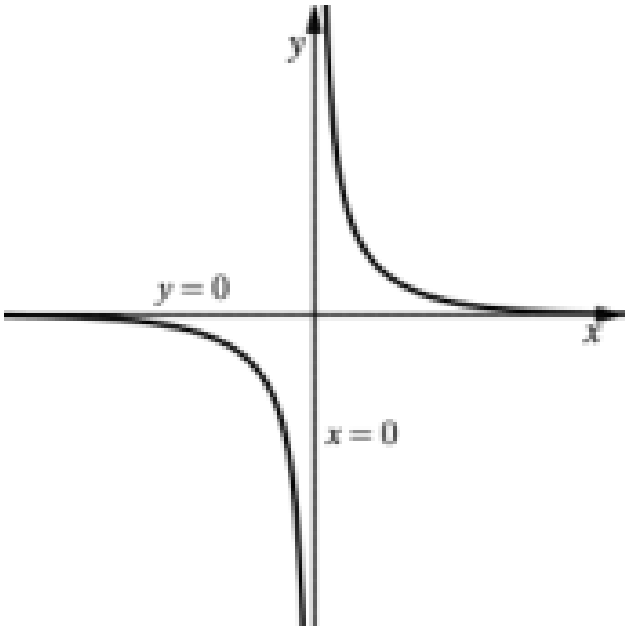
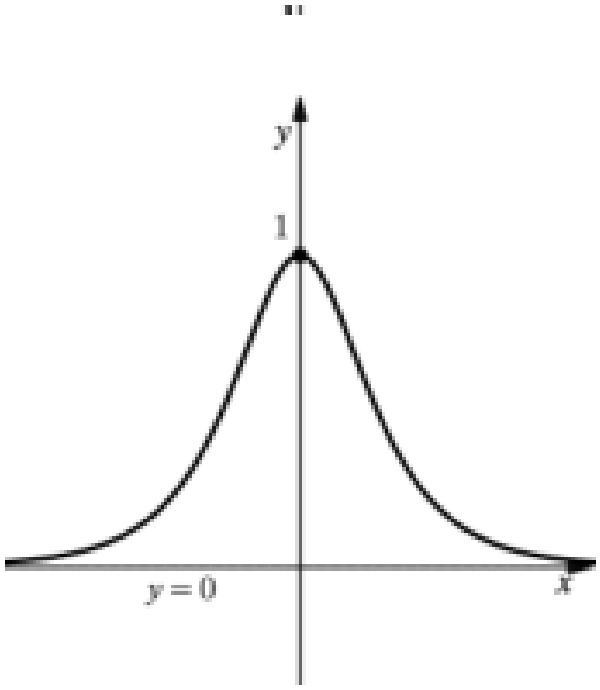
2.18 Graph Sketching: Reciprocal Hyperbolic Functions

Worked Solutions

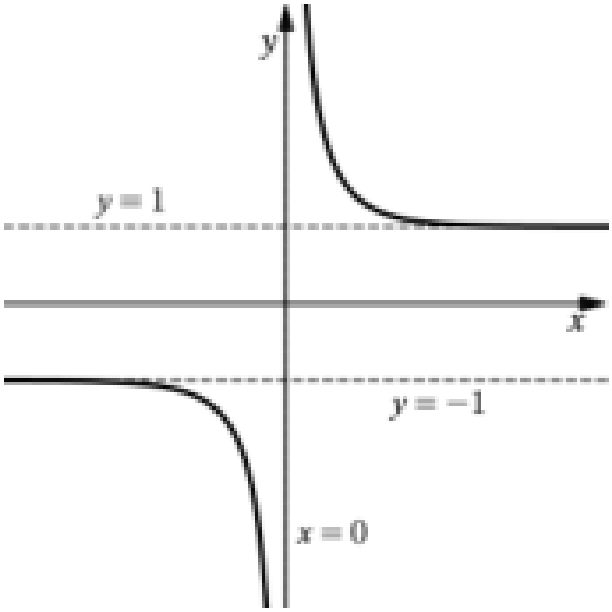
1 a)



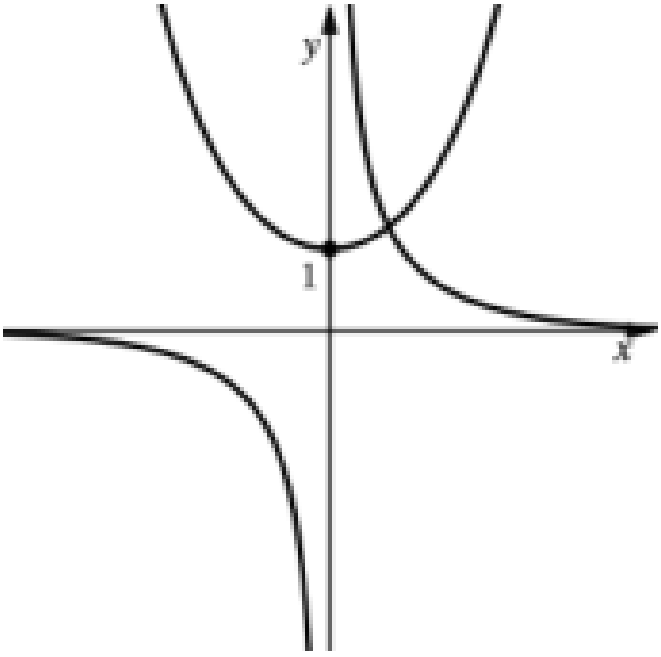
1 b)



1 c)



2 a)



2 b)

$$\cosh x = \operatorname{cosech} x$$

$$\Rightarrow \cosh x = \frac{1}{\sinh x}$$

$$\begin{aligned}
&\Rightarrow \sinh x \cosh x = 1 \\
&\Rightarrow \frac{1}{2}\sinh 2x = 1 \\
&\Rightarrow \sinh 2x = 2 \\
&\Rightarrow 2x = \ln(2 + \sqrt{2^2 + 1}) \\
&\Rightarrow x = \frac{1}{2}\ln(2 + \sqrt{5}).
\end{aligned}$$

Alternatively:

$$\begin{aligned}
\cosh x &= \operatorname{cosech} x \\
\Rightarrow \cosh x &= \frac{1}{\sinh x} \\
\Rightarrow \sinh x \cosh x &= 1 \\
\Rightarrow \frac{e^x - e^{-x}}{2} \times \frac{e^x + e^{-x}}{2} &= 1 \\
\Rightarrow (e^x - e^{-x})(e^x + e^{-x}) &= 4 \\
\Rightarrow e^{2x} - e^{-2x} &= 4 \\
\Rightarrow e^{2x} - e^{-2x} - 4 &= 0 \\
\Rightarrow e^{4x} - 1 - 4e^{2x} &= 0 \\
\Rightarrow e^{4x} - 4e^{2x} - 1 &= 0 \\
\Rightarrow (e^{2x} - 2)^2 - 4 - 1 &= 0 \\
\Rightarrow (e^{2x} - 2)^2 &= 5 \\
\Rightarrow e^{2x} - 2 &= \sqrt{5}.
\end{aligned}$$

Note that $e^{2x} - 2 \neq -\sqrt{5}$ as $e^{2x} - 2 > -2 > -\sqrt{5}$.

$$\begin{aligned}
\therefore e^{2x} &= 2 + \sqrt{5} \\
\Rightarrow 2x &= \ln(2 + \sqrt{5}) \\
\Rightarrow x &= \frac{1}{2}\ln(2 + \sqrt{5}).
\end{aligned}$$

3 a)

The curve must be of the form $y = \operatorname{sech} x + b$. The curve passes through $(\ln(2 + \sqrt{3}), 0)$:

$$\begin{aligned}
0 &= \operatorname{sech}(\ln(2 + \sqrt{3})) + b \\
\Rightarrow -b &= \frac{1}{\cosh(\ln(2 + \sqrt{3}))} \\
\Rightarrow \cosh(\ln(2 + \sqrt{3})) &= -\frac{1}{b} \\
\Rightarrow \frac{e^{\ln(2+\sqrt{3})} + e^{-\ln(2+\sqrt{3})}}{2} &= -\frac{1}{b} \\
\Rightarrow \frac{(2 + \sqrt{3}) + (2 + \sqrt{3})^{-1}}{2} &= -\frac{1}{b} \\
\Rightarrow \frac{(2 + \sqrt{3}) + (2 - \sqrt{3})}{2} &= -\frac{1}{b} \\
\Rightarrow 2 &= -\frac{1}{b} \\
\Rightarrow b &= -\frac{1}{2}.
\end{aligned}$$

Therefore, $y = \operatorname{sech} x - \frac{1}{2}$.

3 b)

The curve must be of the form $y = \operatorname{sech}(x - a)$. The curve passes through $\left(0, \frac{2e^3}{1+e^6}\right)$:

$$\begin{aligned}
\frac{2e^3}{1+e^6} &= \operatorname{sech}(-a) = \operatorname{sech} a \\
\Rightarrow \frac{2e^3}{1+e^6} &= \frac{1}{\cosh a} \\
\Rightarrow \cosh a &= \frac{1+e^6}{2e^3} \\
\Rightarrow a &= \pm \ln\left(\frac{1+e^6}{2e^3} + \sqrt{\left(\frac{1+e^6}{2e^3}\right)^2 - 1}\right) \\
&= \pm \ln\left(\frac{1+e^6}{2e^3} + \sqrt{\frac{e^{12} - 2e^6 + 1}{4e^6}}\right) \\
&= \pm \ln\left(\frac{1+e^6}{2e^3} + \frac{e^6 - 1}{2e^3}\right) \\
&= \pm \ln(e^3) = \pm 3.
\end{aligned}$$

The curve has clearly been translated to the left, so $a = -3$. Therefore, $y = \operatorname{sech}(x + 3)$.

4 a)

$x = -1,\ y = 2,\ y = 0.$

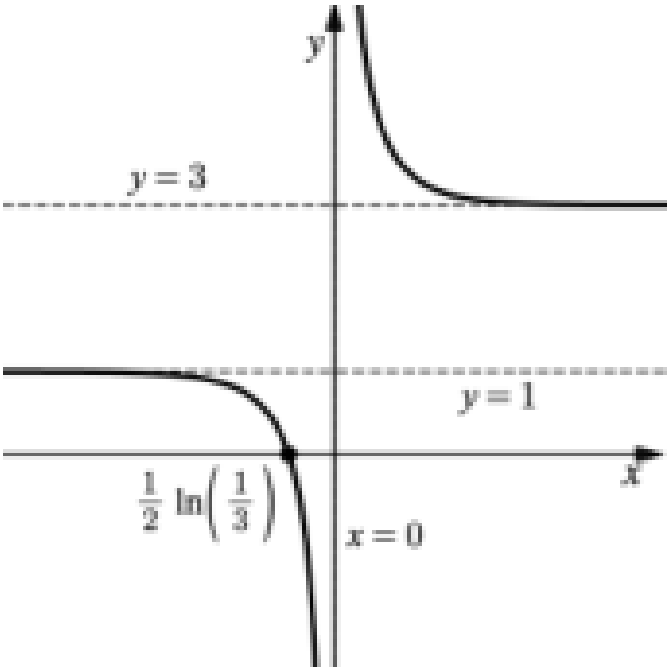
4 b)

$x = 2,\ y = -2,\ y = -4.$

5 a)

When $y = 0$:

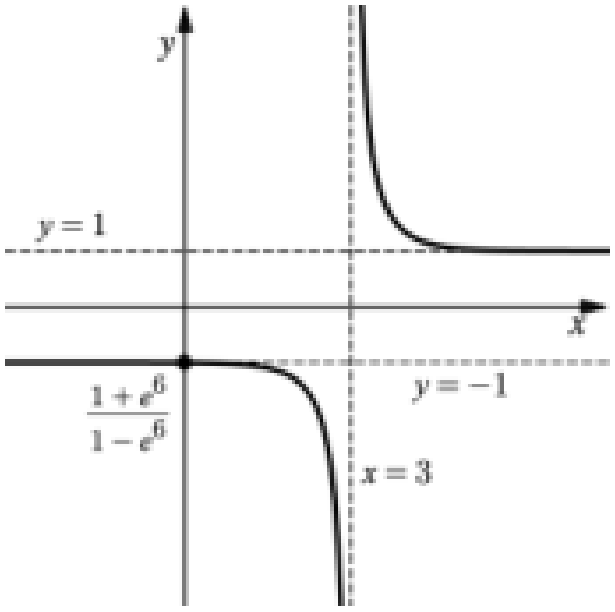
$$\begin{aligned}
&0 = \coth x + 2 \Rightarrow \coth x = -2 \Rightarrow \tanh x = -\frac{1}{2}, \\
&x = \frac{1}{2} \ln\left(\frac{1 - \frac{1}{2}}{1 + \frac{1}{2}}\right) = \frac{1}{2} \ln \frac{1}{3}.
\end{aligned}$$



5 b)

When $x = 0$:

$$y = \coth(-3) = \frac{\cosh(-3)}{\sinh(-3)}$$
$$= \frac{e^{-3} + e^{3}}{e^{-3} - e^{3}} = \frac{1 + e^{6}}{1 - e^{6}} \approx -1.005.$$

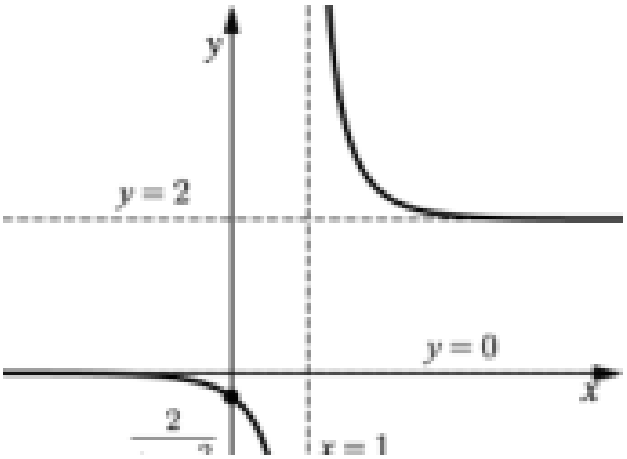


5 c)

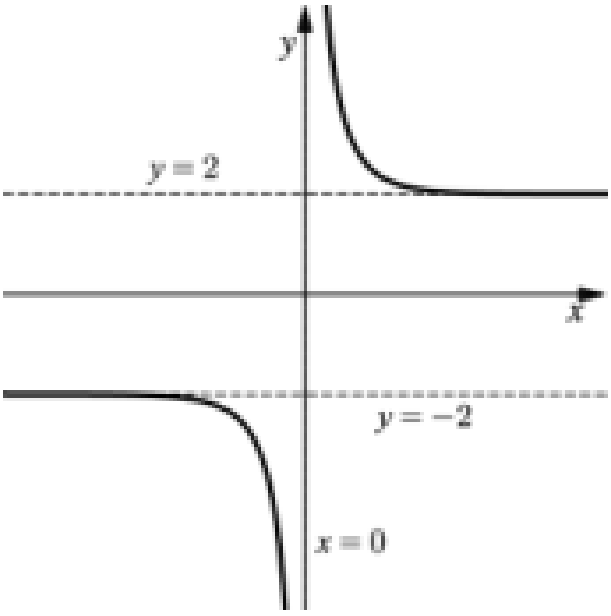
When $x = 0$:

$$y = \coth(-1) + 1 = \frac{e^{-1} + e^{1}}{e^{-1} - e^{1}} + 1$$
$$= \frac{1 + e^{2}}{1 - e^{2}} + 1 = \frac{2}{1 - e^{2}} \approx -0.313.$$

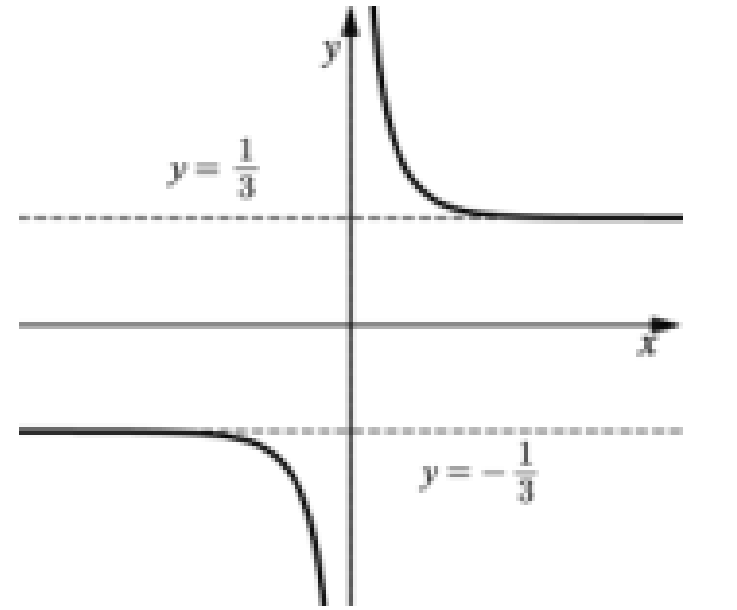
$$\frac{)}{)} + 1 = \frac{\frac{e^{-1}+e^{1}}{2}}{\frac{e^{-1}-e^{1}}{2}} + 1 = \frac{e^{-1}+e^{1}}{e^{-1}-e^{1}} + 1 = \frac{1\cdot}{1\cdot}$$



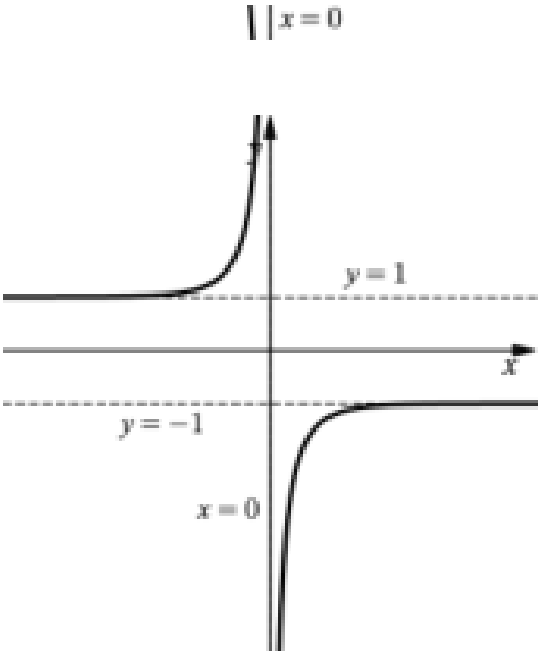
6 a)



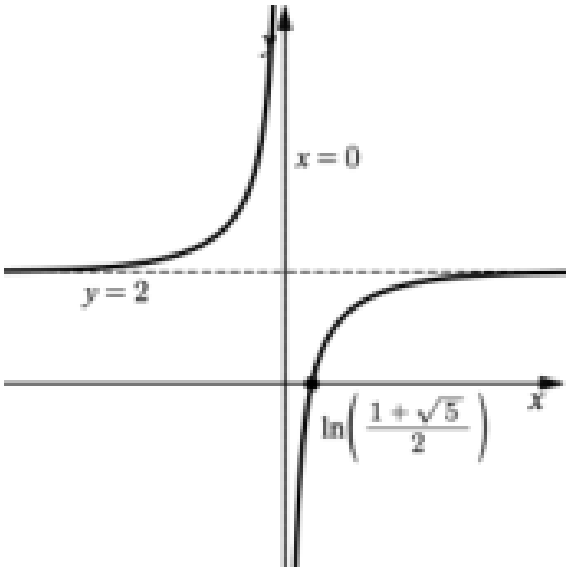
6 b)



6 c)

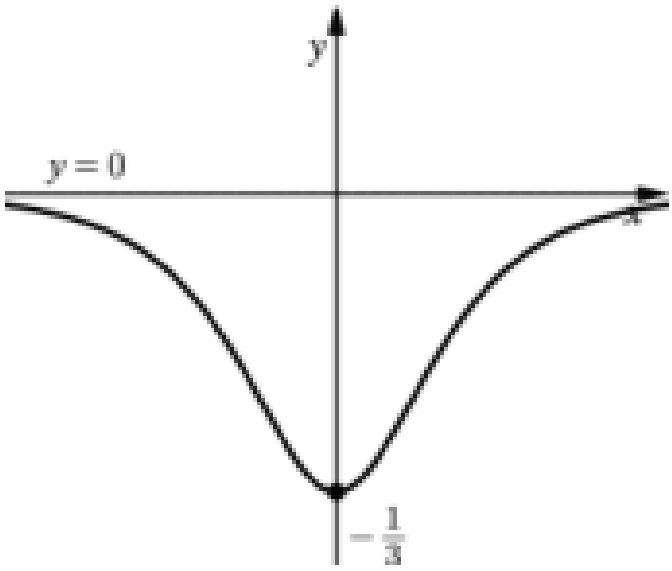


7 a)

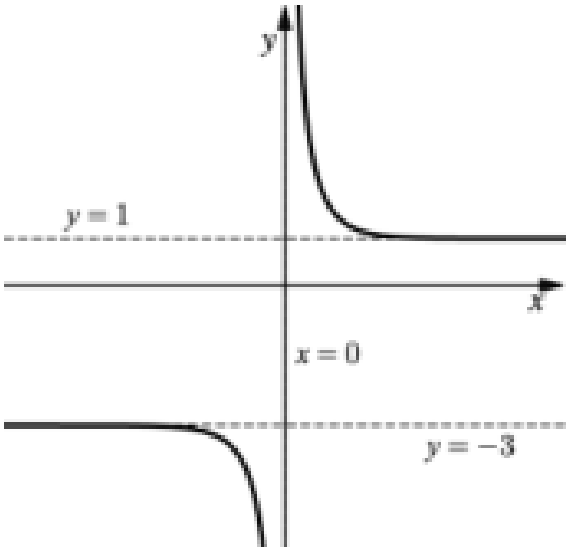


7 b)

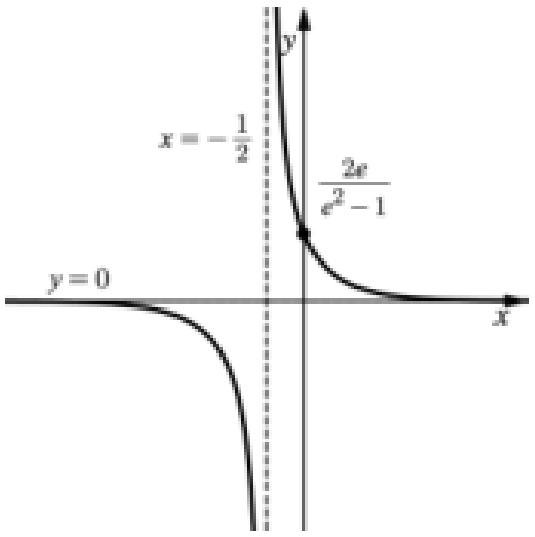
fig. Reciprocal Hyperbolic Functions (wi



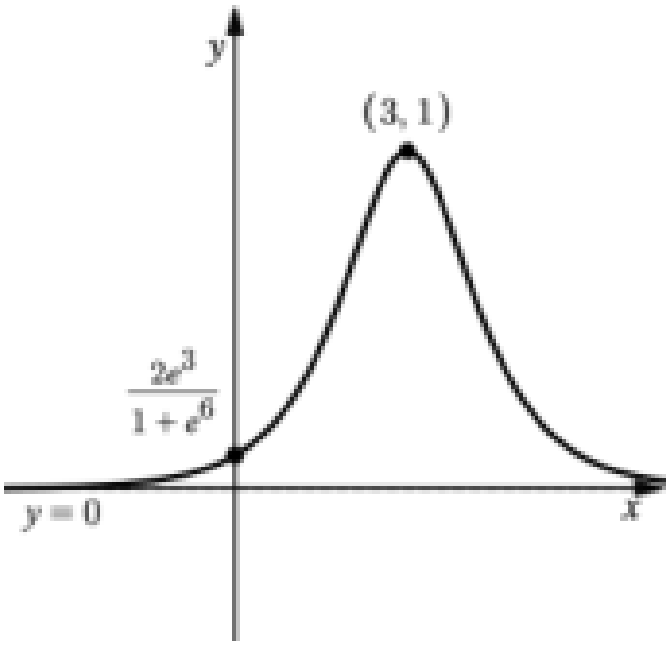
7 c)



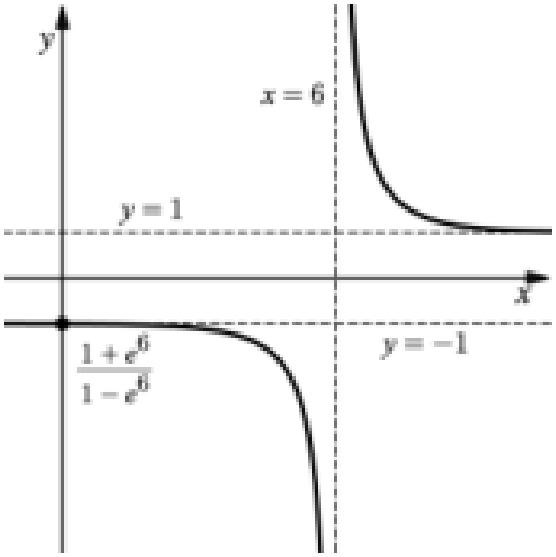
7 d)



7 e)



7 f)



8)

It appears to be a transformation of the curve $y = \coth x$, including a reflection in either the x or y axes. We know that $\coth x$ is an odd function, where $\coth(-x) = -\coth x$, so we can focus our attention on a reflection in the y-axis without causing problems.

The curve has horizontal asymptotes at $y = 1$ and $y = -1$, so no vertical stretches or translations have been applied. The curve has a vertical asymptote at $x = 8$, which might suggest $y = \coth(-x-8)$. However, there may have been a horizontal stretch applied, so we must consider $y = \coth(ax+b)$,

where we will expect a to be a negative constant. Because of the vertical asymptote at $x = 8$, we must have $8a + b = 0$.

The curve passes through $\left(0, \frac{e^8+1}{e^8-1}\right)$, so

$$\begin{aligned}
\frac{e^8+1}{e^8-1} &= \coth b \\
\Rightarrow \frac{e^8+1}{e^8-1} &= \frac{1}{\tanh b} \\
\Rightarrow \tanh b &= \frac{e^8-1}{e^8+1} \\
\Rightarrow b &= \frac{1}{2}\ln\left(\frac{1+\frac{e^8-1}{e^8+1}}{1-\frac{e^8-1}{e^8+1}}\right) \\
&= \frac{1}{2}\ln\left(\frac{(e^8+1)+(e^8-1)}{(e^8+1)-(e^8-1)}\right) \\
&= \frac{1}{2}\ln\left(\frac{2e^8}{2}\right) = \frac{1}{2}\ln(e^8) = 4.
\end{aligned}$$

If $b = 4$, then $8a + 4 = 0 \Rightarrow a = -\frac{1}{2}$. Therefore,

$$y = \coth\left(-\frac{1}{2}x + 4\right).$$